

Figure 26.31 Output buffer used in the op-amp of Fig. 26.29.

Figure 26.32 shows one solution to this problem. We've added a device to cascode the output buffer's NMOS (the one connected to the output terminal). The speed shouldn't be affected by the addition of the device (which operates near or in the triode region). The added device allows the op-amp's output to swing more freely (but the current still won't be precisely set). The added device won't affect the output swing range of the op-amp. Figure 26.33 shows the op-amp with modified output buffer.

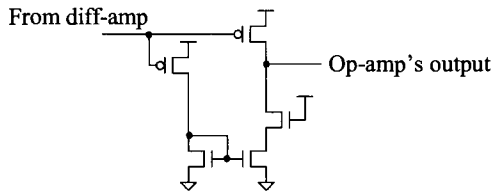


Figure 26.32 Adding a device to allow the output voltage to swing.

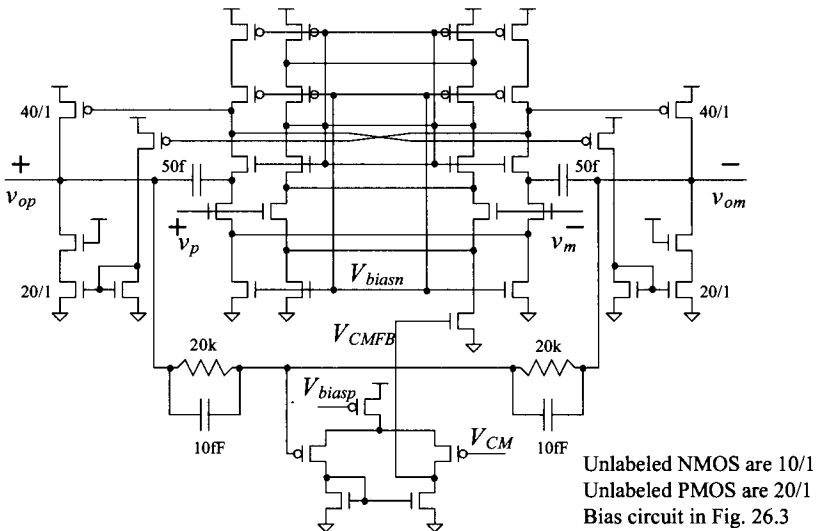


Figure 26.33 Op-amp with modified output buffer.

Simulation Results

Figures 26.34 to 26.36 show simulation results based on the circuit topologies in Figs. 26.23 to 26.25, respectively, using the op-amp in Fig. 26.33. In Fig. 26.34 we see that the outputs now swing close to the power supply rails (unlike what we saw in Fig. 26.30). Also, the gain is higher. Figure 26.35 shows the step response of the amplifier in the topology seen in Fig. 26.24. Finally, Fig. 26.36 shows the outputs of the sample-and-hold in Fig. 26.25 using the op-amp in Fig. 26.33.

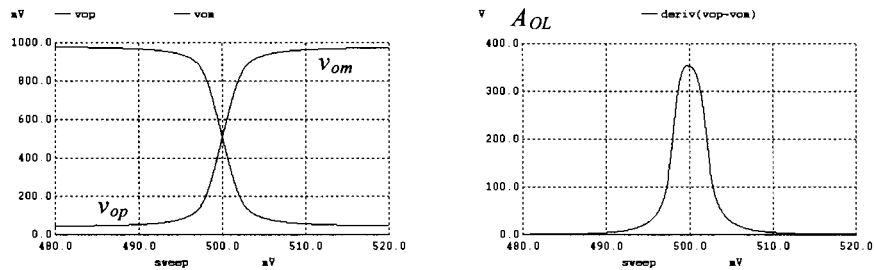


Figure 26.34 Resimulating the op-amp in Fig. 26.33 in the configuration seen in Fig. 26.23.

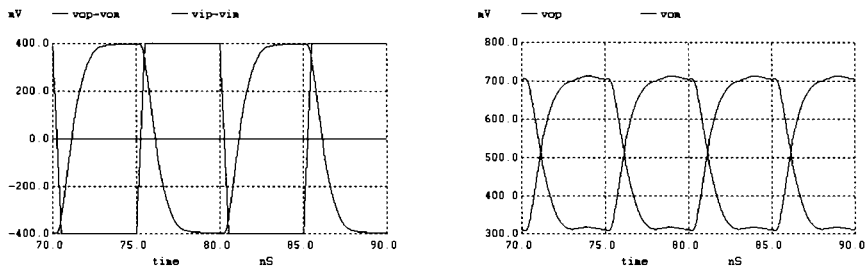


Figure 26.35 Regenerating the simulation results using the topology in Fig. 26.24 with the op-amp in Fig. 26.33.

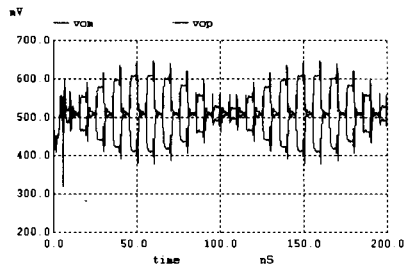


Figure 26.36 Using the op-amp in Fig. 26.33 in the sample-and-hold circuit of Fig. 26.25. Figure shows the outputs of the op-amp.

Using MOSFETs Operating in the Triode Region

For the output buffer in Fig. 26.32, we added a transistor in the drain portion of the circuit. We might wonder if we can accomplish better control of the current flowing in the output buffer by adding circuitry to the source side of the buffer. Towards answering this question, consider the portion of the output buffer seen in Fig. 26.37. If both outputs are at V_{CM} , the gate-source voltages of M2 and M3 are the same and so are the gate-source voltages of M1 and M4. If one output goes high and the other output goes low (and the outputs are centered around V_{CM}), then the net drain current of each M3 is constant. If one output goes below the threshold voltage of an NMOS device, then the balancing stops working. The practical problem of using triode-operating MOSFETs (in a CMFB circuit or any type of amplifying configuration) is that the gain through a MOSFET operating in the triode region is low (so it's difficult to provide control).

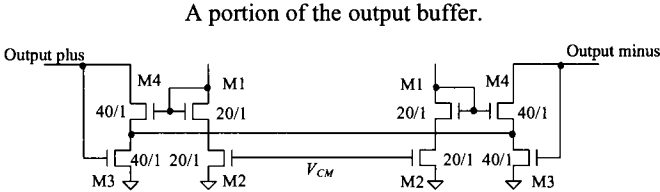


Figure 26.37 Using triode-operating MOSFETs to balance the outputs (bad).

Start-up Problems

Examine the circuit in Fig. 26.38 using the op-amp in Fig. 26.33. Because the op-amp's inputs are at 0 V, the diff-amps on the input of the op-amp are off. This causes the gates of the PMOS devices in the output buffer to be pulled to V_{DD} . The circuit remains in this state and doesn't move the outputs or the inputs up to V_{CM} . To avoid this start-up problem, we need to ensure that there is some DC path to V_{DD} or V_{CM} to "start-up" the op-amp.

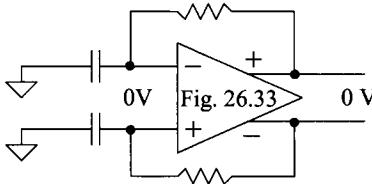


Figure 26.38 The op-amp in Fig. 26.33 won't turn on in this topology.

Lowering Input Capacitance

Consider connecting the gates of the NMOS diff-amp used for biasing to V_{CM} (see the bold line in Fig. 26.39). This reduces the input capacitance of the op-amp and shouldn't affect the normal operation as long as the common-mode voltage of the op-amp (the gates

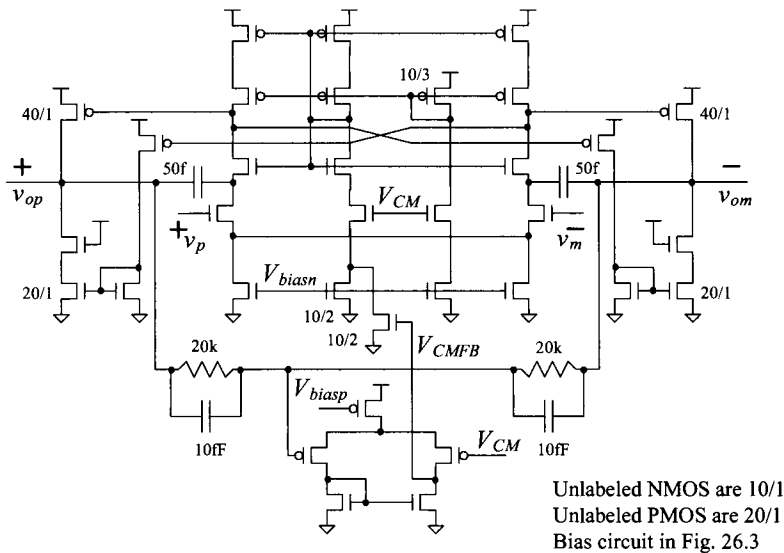


Figure 26.40 Making the op-amp more practical.

A more practical problem is the change in the MOSFET's drain currents with drain-source voltages. For example, looking at Figs. 26.18a and b, we see that it's possible for the NMOS's drain current to be twice the PMOS's drain current for the same gate-source voltages. This can cause op-amp failure in the CMFB circuit. If, for example, the current flowing in the 10/2 MOSFET connected to V_{biasn} is larger than the current sourced by the PMOS, the voltage V_{CMFB} goes to zero and the CMFB loop doesn't work properly. A good "rule-of-thumb" is for the fixed current flowing in the CMFB-controlled bias circuit to be 25–50% of the total expected current.

Increasing the Op-Amp's Open-Loop Gain

The op-amp that we've developed in this section has an open-loop gain in the hundreds. As we'll see in Ch. 29 (Eq. [29.59]), the open-loop gain of the op-amps used in a data converter has a direct effect on the maximum attainable resolution. Towards increasing the gain of the op-amp, let's use the gain-enhancement (GE) techniques presented in Sec. 24.4.

Figure 26.41 shows how we should *not* implement GE. An amplifier with fully-differential outputs, like the amplifier in Fig. 26.7, regulates the drains of the top PMOS devices. We know that this is bad because we would need a CMFB circuit to balance the outputs of the added amplifier.

Figure 26.42 shows how we can add GE to the op-amp developed in this section. Notice that GE is implemented with amplifiers having single-ended outputs. When designing the added amplifiers, as discussed in Sec. 24.4, the bandwidth isn't important for high-speed operation. The lengths of the MOSFET in the added amplifiers can be increased to reduce power and boost gain. An important concern is the added amplifier's input common-mode range.

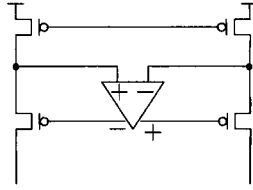


Figure 26.41 How not to implement GE in an op-amp (unless the added amplifier employs CMFB).

The practical problem with the topology seen in Fig. 26.42 is the implementation of the CMFB. With the GE added to the op-amp we now have four additional feedback loops. Variations in V_{CMFB} affect all GE loops in addition to the CMFB through the op-amp. Making these loops stable becomes extremely challenging. What we need is to implement the CMFB without including the diff-amp and GE amplifiers. We can only do this by controlling the output buffer common-mode level, Fig. 26.43. This circuit includes the output buffer we used in Fig. 26.37. Now, however, we add an amplifier in series with the triode-operating MOSFETs to boost the CMFB gain (so that the outputs can be balanced around V_{CM}). The problem with this approach is that the CMFB isn't compensated using the same capacitors as the differential forward signal path (i.e., no Miller effect). We must be concerned with CMFB stability.

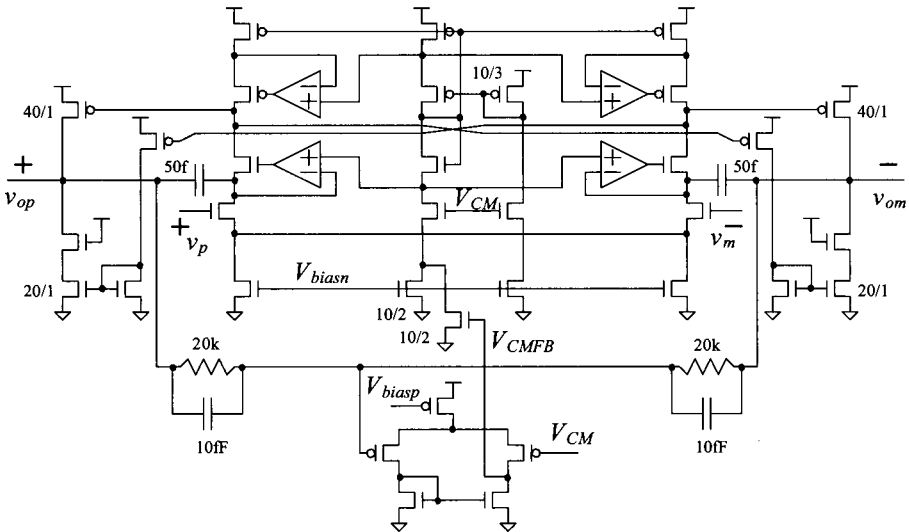


Figure 26.42 Adding gain-enhancement to the op-amp.

Let's discuss the op-amp design in Fig. 26.43. We begin by showing the DC behavior of the op-amp (to show that the outputs are indeed balanced). Figures 26.44a and (b) show the DC behavior and gain of the op-amp in Fig. 26.43 in the topology seen in Fig. 26.24. Again note that by increasing the values of the 20k resistors used to average

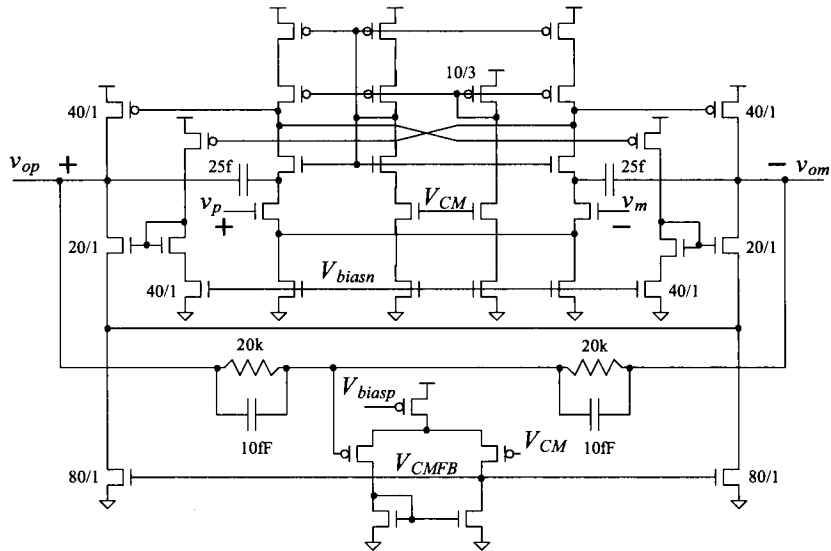


Figure 26.43 Providing CMFB through just the output buffer. Using an amplifier with triode-operating MOSFETs for CMFB (good).

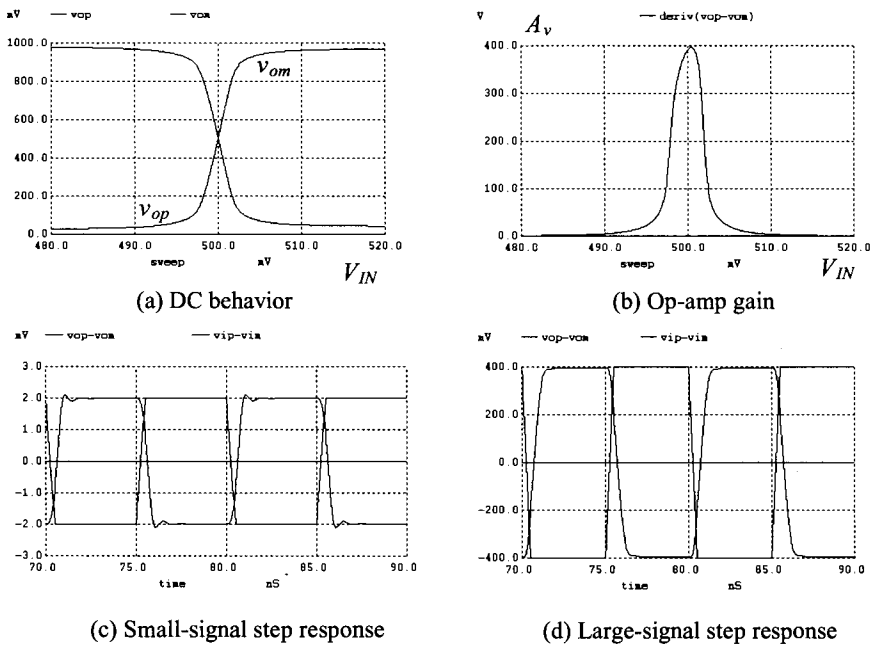


Figure 26.44 Behavior of the op-amp in Fig. 26.43 (see text).

the outputs we can increase the DC gain of the op-amp. Because the forward differential path compensation (now) doesn't include the CMFB path, we've reduced the compensation capacitors from 50 fF to 25 fF (there are practical issues with this change that we'll discuss next). Note that the widths of the added triode-operating MOSFETs are increased to ensure that they operate well within the triode region and don't significantly affect the output drive capability of the op-amp. Figures 26.44c and (d) show the small- and large-signal step responses (see Fig. 26.24). The settling times are approximately 1–2 ns. The current drawn from V_{DD} (including the the bias circuit current) is approximately 250 μA (200 μA for the op-amp alone).

Offsets

We said in Sec. 26.2 that we want to add offsets of 50 mV in series with the gates of our MOSFETs to see if the op-amp “breaks.” Consider the addition of an input-referred offset voltage in the schematic seen in Fig. 26.45. This offset is used to model the overall offset voltage of the op-amp (see Fig. 24.4). If we look at the effects of the offset on the circuitry in Fig. 26.43, we see that one of the NMOS devices in the diff-pair will have a higher overdrive voltage (larger g_m) than the other NMOS device. Since the unity-gain frequency of an op-amp is given by $g_m/2\pi C_c$, the effect is a shift in the unity-gain frequency (and potential instability). As seen in Fig. 26.45, using the 25 fF compensation capacitors destabilizes the op-amp (as indicated by the ringing). Increasing the compensation capacitor's value (back) to 50 fF makes the step response cleaner. We could argue that 50 mV is an unrealistically high offset voltage, so it's better to leave the capacitors at 25 fF (and this may be the case). However, in a practical CMOS process, the op-amp's characteristics shift with the process shifts (and temperature). It's better to overcompensate than to have an unstable op-amp.

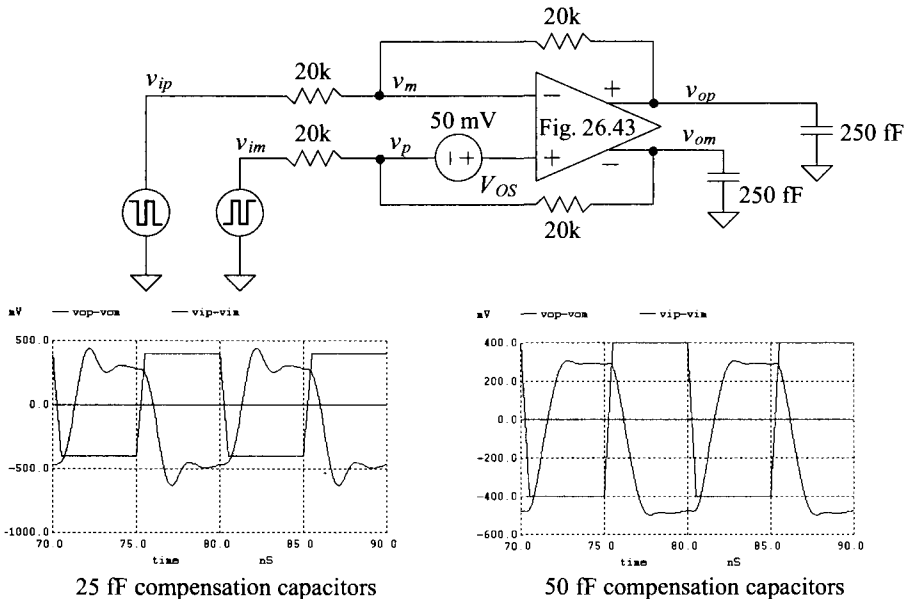


Figure 26.45 How an offset can affect the step response (compensation).

Op-Amp Offset Effects on Outputs

Notice, in Fig. 26.45, that an offset shifts the differential output signal by twice the offset voltage. The individual op-amp output voltages remain centered around V_{CM} . To describe this in more detail, consider the test setup seen in Fig. 26.46. We know the CMFB forces

$$\frac{v_{op} + v_{om}}{2} = V_{CM} \text{ or } v_{op} = 2V_{CM} - v_{om} \quad (26.8)$$

Further, assuming large op-amp open-loop gain,

$$v_p + V_{OS} = v_{pos} \approx v_m \quad (26.9)$$

Equating currents, we can write

$$\frac{V_{CMI} - v_p}{R_{in}} = \frac{v_p - v_{om}}{R_f} \text{ or } \frac{V_{CMI} - v_m + V_{OS}}{R_{in}} = \frac{v_m - V_{OS} - v_{om}}{R_f} \quad (26.10)$$

and

$$\frac{V_{CMI} - v_m}{R_{in}} = \frac{v_m - v_{op}}{R_f} \quad (26.11)$$

Subtracting Eq. (26.11) from Eq. (26.10), we get

$$\frac{V_{OS}}{R_{in}} = \frac{-V_{OS} + v_{op} - v_{om}}{R_f} \quad (26.12)$$

or

$$v_{op} - v_{om} = V_{OS} \cdot \left(\frac{R_f}{R_{in}} + 1 \right) \quad (26.13)$$

The voltages on the outputs of the op-amp are then

$$v_{op} = \frac{V_{OS}}{2} \cdot \left(1 + \frac{R_f}{R_{in}} \right) + V_{CM} \text{ and } v_{om} = V_{CM} - \frac{V_{OS}}{2} \cdot \left(1 + \frac{R_f}{R_{in}} \right) \quad (26.14)$$

The op-amp's input voltages are

$$v_m \approx v_p + V_{OS} = \frac{V_{OS}}{2} + V_{CM} \cdot \left(\frac{R_{in}}{R_{in} + R_f} \right) + V_{CMI} \cdot \left(\frac{R_f}{R_{in} + R_f} \right) \quad (26.15)$$

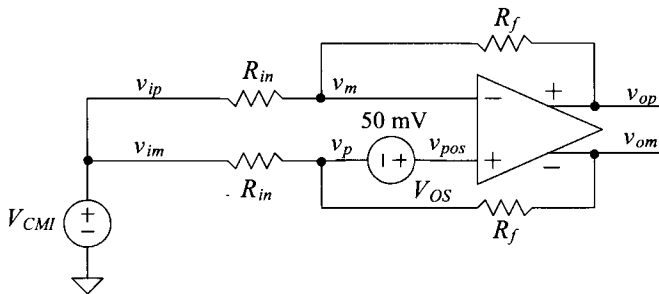


Figure 26.46 How an offset voltage causes an imbalance in the outputs.

If the input signal's common-mode voltage, V_{CMI} , is the same as the op-amp's common-mode voltage, V_{CM} , and V_{OS} is zero, then both op-amp inputs are held at V_{CM} . If $V_{CM} = V_{CMI}$ but the offset isn't zero, then $v_m = V_{OS}/2 + V_{CM}$ and $v_p = V_{CM} - V_{OS}/2$. Neglecting the offset voltage, if the input signal's common mode voltage isn't V_{CM} , then the op-amp's input common-mode voltage ($v_p = v_m$) will be different from the ideal value of V_{CM} . This can shut the op-amp off and cause undesirable behavior.

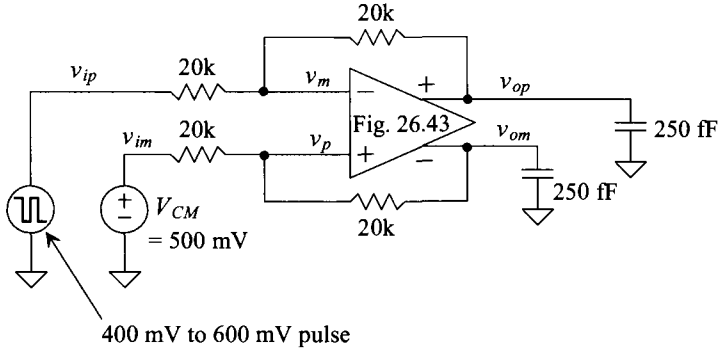


Figure 26.47 Problems with single-ended input signals.

Single-Ended to Differential Conversion

Even if the input signals are referenced around V_{CM} and the op-amp is offset free, we can still have problems. Consider the circuit in Fig. 26.47. In this circuit the input signal is single-ended. The other input to the op-amp is tied to V_{CM} . Even though the input is balanced around 500 mV, it is not a truly differential signal. When the input signal is up at 600 mV, the effective input common-mode voltage, V_{CMI} , is 550 mV (the average of the two inputs). When the input signal is at 400 mV, V_{CMI} is 450 mV. Figure 26.48 shows the simulation results using the circuit in Fig. 26.47. These results are very interesting. Consider what's happening between 70 and 75 ns. During this time, v_{ip} is 600 mV and v_{im} is 500 mV (and so $v_{ip} - v_{im} = 100$ mV). The op-amp's inputs move above the ideal 500 mV, as seen in the figure. This has the effect of increasing the transconductance of the diff-amp (the diff-amp's tail current is operating near/in the triode region so the tail

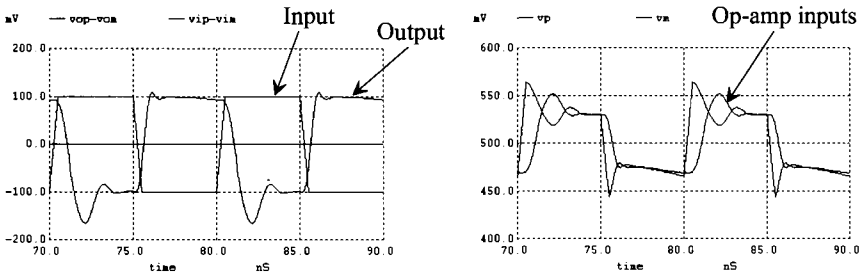


Figure 26.48 Simulating the operation of the circuit in Fig. 26.47.

current increases as the input common-mode voltage goes up). The increase in the diff-amp's g_m causes the unity gain frequency to increase and, with a fixed compensation capacitor, the phase margin to decrease (causing overshoot and ringing in the circuit's step response). Next consider what happens between 75 and 80 ns. The input signal switches to $400\text{ mV} - 500\text{ mV} = -100\text{ mV}$ and the op-amp's inputs drop down to around 475 mV. This causes the diff-amp's g_m to decrease and the op-amp to slow down (become more stable). However, at such a low input common-mode voltage the gain of the op-amp will drop and the outputs may wander. (See Sec. 30.3.1 for design information).

CMFB Settling Time

If we look at the voltage, V_{CMFB} , in the op-amps in Fig. 26.40 or 26.43, we may see that they aren't settling as quickly as the differential mode signals. Is this bad? As seen in Eq. (26.13), variations in V_{CM} don't (ideally) affect the differential output signal. However, if the variations in V_{CMFB} and thus V_{CM} are large, one of the output signals can saturate at close to V_{DD} or ground (the op-amp's gain drops), affecting the differential signal path.

CMFB in the Output Buffer (Fig. 26.43) or the Diff-Amp (Fig. 26.40)?

In this section we've presented the design of op-amps using continuous-time CMFB. Various design trade-offs and topologies were presented. At this point a good question is: "Which CMFB scheme is better?" Controlling the output common-mode level through the output buffer (Fig. 26.43) is simple, easy to ensure stability, and fairly robust. However, consider what happens if the op-amp in Fig. 26.43 is used in the topology in Fig. 26.46 (without an offset). If V_{CM} is ground, then, according to Eq. (26.15), the input voltages to the op-amp will move to 250 mV while the outputs of the op-amp remain at 500 mV. Thinking about this for a moment and neglecting the fact that an op-amp input common-mode voltage of 250 mV will shut the op-amp off, we see that the outputs of the op-amp must source a current back through the feedback resistors to the inputs. Further, anytime $V_{CM} < V_{CM}$, the PMOS in the output buffer must source a DC current back to the inputs. Similarly, if $V_{CM} > V_{CM}$, the NMOS in the output buffer must sink a current from the inputs of the op-amp. When using CMFB in the output buffers, we have no way of increasing the quiescent or DC current flowing in the output buffer to source/sink current from the input source (or to a DC load connected to ground). The output voltage of the diff-amp in the op-amp of Fig. 26.43 biases, or sets, the quiescent current in the output buffer. The CMFB scheme used in the output buffer simply adjusts the drive strength of the NMOS (in the output buffer) to set the op-amp's output common-mode level.

Using the CMFB scheme in Fig. 26.40, we can adjust the output voltage of the diff-amp (and ultimately the op-amp's output voltage) and thus the bias current in the output buffer (both the NMOS and the PMOS connected to the output buffer). However, if the output buffer must source/sink a significant amount of current, we can have problems with this topology too. For example, if the diff-amp's output voltage increases (turning off the PMOS in the output buffer and attempting to pull the op-amp's outputs down) while the NMOS devices connected to the outputs are sinking significant current, then it's possible that the outputs will get pulled upwards and the CMFB will fail (likely causing the gain of the op-amp to drop and the outputs to have limited swing).

To ensure the most robust op-amp design (biasing tolerant to offsets and easy to compensate the CMFB loops), CMFB circuits can be placed around both op-amp stages. This method is used in our last op-amp design example discussed next.